Reichman University

Text Retrieval and Search Engines

Lecture 12
Positional Language Models

Semester 1 - 2026

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  - Term proximity heuristic
  - Passage retrieval

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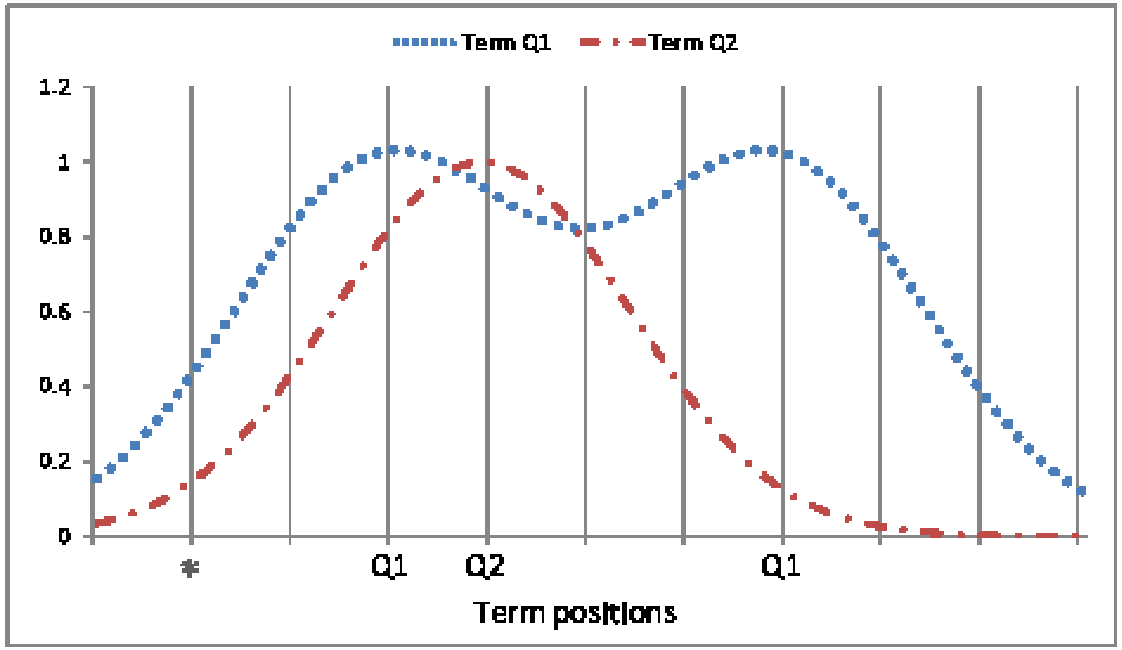
- **Open questions:**
  - How to define the propagation function?
  - How to estimate the PLM?
  - How to score the document?

Term Propagation



- PLM at each position can be estimated based on all propagated counts of all words to the position
- Treat a position as a document with all words appearing at that position with discounted counts

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Virtual document at position i

$\langle c(w_1, i), \ldots, c(w_V, i) \rangle$

Smoothing

- PLM at position i (unsmoothed)
  - $P_{MLE}(w|M_{d,i}) = \frac{c(w,i)}{\sum_{w' \in V} c(w',i)}$
- Jelinek-Mercer (JM) smoothing

$$p_{JM}(w|M_{d,i}) = (1-\lambda)p_{MLE}(w|M_{d,i}) + \lambda p_{MLE}(w|C)$$

- Dirichlet smoothing

$$p_{Dir}(w|M_{d,i}) = \frac{c(w,i) + \mu p_{MLE}(w|C)}{Z_i + \mu}$$

  - $Z_i = \sum_{w \in V} c(w,i)$ – the length of the virtual document at position i

Similarity Measure

- To score a position given a query we can use the KL-divergence

$$\mathrm{S}(\mathrm{q},\mathrm{d},\mathrm{i}) = -KL\left(M_q \middle| M_{d,i}\right) = -\sum_{w\in V} p\left(w \middle| M_q\right) log \frac{p\left(w \middle| M_q\right)}{p\left(w \middle| M_{d,i}\right)}$$

Where,

- V is the vocabulary
- $p(w|M_q)$ can be estimated using:
  - MLE
  - Pseudo relevance feedback algorithms – e.g., relevance model

Propagation Functions - $k(i,j)$

- Gaussian kernel

$$k(i,j) = \exp[\frac{-(i-j)^2}{2\sigma^2}]$$

- Triangle kernel

$$k(i,j) = \begin{cases} 1 - \frac{|i-j|}{\sigma} & if\ |i-j| \leq \sigma \\ 0 & otherwise \end{cases}$$

- Passage kernel (non-proximity-based kernel)

$$k(i,j) = \begin{cases} 1 & if\ |i-j| \leq \sigma \\ 0 & otherwise \end{cases}$$

- σ – the steepness parameter
  - “soft” passage representation

Analysis of the Steepness Parameter

- For $\sigma \rightarrow \infty$: $k(i,j) = 1$ for all i and j

$$c(w,i) = \sum_{j=1}^{|d|} I(w,j)k(i,j) = c(w,d)$$

Standard whole document language model ($p(w|M_d)$) is a private case of PLM

- Small σ emphasizes more local term proximity

PLM-Based Document Ranking

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- **Multi-σ Strategy**
  - Interpolation of the PLM and the regular document language model

$$S_{multi-\boldsymbol{\sigma}}(q,d) \equiv \gamma * max_{i\in[1,N]}\{S(q,d,i)\} + (1-\gamma) * (-KL(M_q|M_d))$$

Class Exercise

- Compute the similarity between the PLM induced at position 1 in document D2 with respect to the query
  q = "onion soup onion"
- Use DIR smoothing with μ=100 and MLE for the query model
- For the propagation function use the Gaussian kernel with σ=5

D1: onion vegetable soup vegetable mushroom
D2: corn onion soup
D3: potato pumpkin tofu potato tofu potato

term	onion	vegetable	soup	mushroom	corn	potato	pumpkin	tofu
tf	2	2	2	1	1	3	1	2
P_{MLE}	2/18	2/18	2/18	1/18	1/18	3/18	1/18	2/18

Partial Solution

- D2: corn onion soup; q = onion soup onion
- $p_{Dir}\left(onion \middle| M_{D2,1}\right) = \frac{c(onion,1)+100p_{MLE}(onion|C)}{Z_i+100}$
- $Z_i = \sum_{w \in V} c(w,1) = c(corn,1) + c(onion,1) + c(soup,1)$
  - $c(w,i) = \sum_{j=1}^{|D2|} I(w,j)k(i,j)$
  - $Z_i = \sum_{w \in V} c(w,1) =$

$$\begin{aligned} &1k(1,1) + 0k(1,2) + 0k(1,2) + && \longrightarrow \text{c(corn,1)} \\ &0k(1,1) + 1k(1,2) + 0k(1,3) + && \longrightarrow \text{c(onion,1)} \\ &0k(1,1) + 0k(1,2) + 1k(1,3) = && \longrightarrow \text{c(soup,1)} \\ &k(1,1) + k(1,2) + k(1,3) && \end{aligned}$$

Partial Solution

- $Z_i = \sum_{w \in V} c(w, 1) = 1k(1,1) + 0k(1,2) + 0k(1,2) + 0k(1,1) + 1k(1,2) + 0k(1,3) + 0k(1,1) + 0k(1,2) + 1k(1,3) = k(1,1) + k(1,2) + k(1,3) =$

$$\exp[0] + \exp\left[-\frac{1}{50}\right] + \exp\left[-\frac{4}{50}\right] = 2.9033$$

$$= The\ length\ of\ the\ virtual\ document\ at\ position\ 1$$

Partial Solution

- $p_{Dir}\left(onion\middle|M_{D2,1}\right) = \frac{c(onion,1)+100p_{MLE}(onion|C)}{Z_i+100} = \frac{1k(1,2)+100*\frac{2}{18}}{2.9033+100} = \frac{\exp\left[-\frac{1}{50}\right]+100*\frac{2}{18}}{2.9033+100} = 0.117502$
- $p_{Dir}\left(soup\middle|M_{D2,1}\right) = \frac{c(soup,1)+100p_{MLE}(soup|C)}{Z_i+100} = \frac{1k(1,3)+100*\frac{2}{18}}{2.9033+100} = \frac{\exp\left[-\frac{4}{50}\right]+100*\frac{2}{18}}{2.9033+100} = 0.11695$
- $S(q, D2,1) = -p(onion|q)ln\frac{p(onion|q)}{p(onion|M_{D2,1})} - p(soup|q)ln\frac{p(soup|q)}{p(soup|M_{D2,1})} = -\frac{2}{3}ln\frac{\frac{2}{3}}{0.117502} - \frac{1}{3}ln\frac{\frac{1}{3}}{0.11695} = -1.506355$

Performance

method\data	WT2G	TREC8	FR	AP88-89
KL	0.2931	0.2509	0.2697	0.2196
$\sigma = 25$	0.3247^+	0.2562^+	**0.2936**	$\mathbf{0.2237}^+$
$\sigma = 75$	$\mathbf{0.3336}^+$	0.2553^+	0.2896^+	0.2227
$\sigma = 125$	0.3330^+	0.2559^+	0.2885	0.2201
$\sigma = 175$	0.3324^+	$\mathbf{0.2574}^+$	0.2858	0.2196
$\sigma = 275$	0.3255^+	0.2561^+	0.2852	0.2193

Table 5: The best performance of multi-σ strategy for different σ. '+' means that improvements over the baseline KL method are statistically significant.

Taken from Lv&Zhai '09